Assignment 4: SA + GA

Part 6: Data collection, Visualization and Comparison

Which algorithm converged faster?

Genetic Algorithm converged faster, around 40 iterations vs just under 7500 that it took SA to converge. GA seemed to explore less paths and stopped changing very early. Once SA begins to converge it happens over more iterations than GA that explores very close to the starting length and makes little improvement.

Which showed more variability?

Simulated Annealing had more variability length changed by more where Genetic Algorithm converges in early generations almost right away. SA reports exploring current length that varies from best inconsistently. This is evidence that the algorithm accepted less promising paths on occasion leading to an overall improvement to the solution it started with.

Did SA escape local minima? Evidence?

Yes, current distance is reported as longer than best as some intervals suggesting the algorithm explored less promising paths at times. The best report continues to improve for sometime before it converges suggesting it did find a better path by exploring. This is consistent with temperature (T) decreasing in later iterations to reduce probability of accepting less promising paths.

```
[SA] iter=  1000  T=0.1213  cur=8.7667  best=6.0214
[SA] iter=  1500  T=0.09446  cur=7.9573  best=6.0214
[SA] iter=  2000  T=0.07356  cur=8.3800  best=6.0214
[SA] iter=  2500  T=0.05728  cur=7.3791  best=6.0214
[SA] iter=  3000  T=0.04461  cur=6.7793  best=6.0214
[SA] iter=  3500  T=0.03474  cur=6.5801  best=6.0214
[SA] iter=  4000  T=0.02705  cur=6.5972  best=6.0214
[SA] iter=  4500  T=0.02107  cur=5.9446  best=5.9446
[SA] iter=  5000  T=0.01641  cur=5.5555  best=5.5555
```

Did GA lose diversity? Evidence?

There is support for GA loss of diversity with a mutation rate of 0.2 and elite at 5. The algorithm quickly converges around generation 40-50 and improves very little. With a low mutation rate and higher elitism less paths are explored and the most promising path found early on is favored. Further support for a loss of diversity is found by improving GA.

Part 7: Improving GA

After tuning different parameters and analyzing the behavior of the Genetic Algorithm: I found that increasing the mutation rate from 0.2 to 0.6

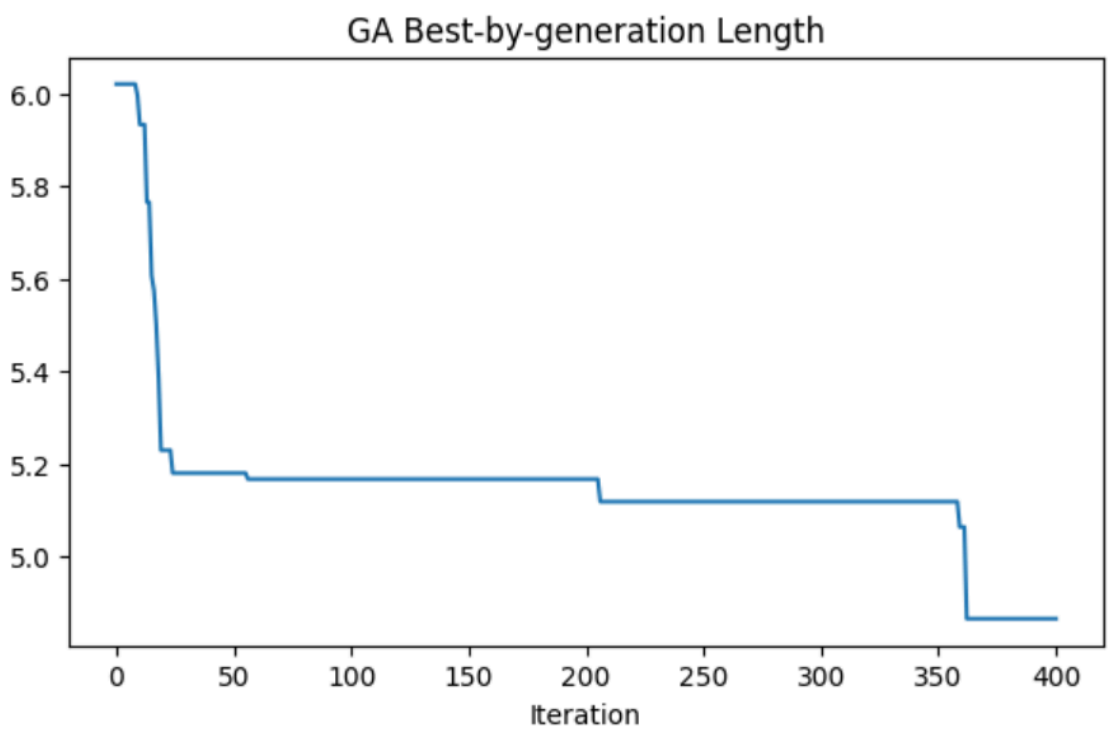
increases its improvement to 17%. However, it does converge in very early generations still. The crossover rate at 0.9 seems to generate the best improvement with current parameters. However, adjusting mutation rate further to 0.8 fixes the early convergence in the 40th generation and now converges much later, around the 300th iteration with a decline in improvement down to 10%. Finally, with a mutation rate of 0.6 and an elitism bias at 3 (reduced from 5) the best length is found with 19.7% improvement and convergence in the later generations.

```
GAConfig(pop_size=200, generations=400, tournament_k=5, crossover_rate=0.9,
          mutation_rate=0.6, elite_size=3, seed=999, report_every=50)
           #mutatoin_rate tuned from 0.2 to create better diversity and impr
our, ga_best_len, ga_hist = genetic_algorithm_tsp(cities, init_seed_tours=[n

A best length: {ga_best_len:.4f}")
A improvement vs NN: {(nn_len - ga_best_len) / nn_len * 100:.2f}%")

(cities, ga_best_tour, title=f"GA Best (L={ga_best_len:.4f})")
ory(ga_hist, title="GA Best-by-generation Length", ylabel="Length")
```

```
GA] gen= 400  best=4.8658
A best length: 4.8658
A improvement vs NN: 19.19%
```



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Disclosure of AI tools:

- Google Colab code autocomplete generation(tab-complete) was used to draft initial versions of several TODO functions. I reviewed the accepted suggestions by comparing my own understanding then validated proper behavior by running the cells and analyzing the outputs. Colab inline autocomplete does not provide a transcript. Inline comments are my own understanding of the code.

- ChatGPT was used to explain concepts, clarify python starter code, and suggest parameter tuning for the Genetic Algorithm. Parameter tuning suggestions by ChatGPT were not fruitful. Actual Genetic Algorithm improvement was achieved when I located the proper section of the notebook to adjust the mutation rate and increase the number of printed results to analyze behavior. Log is saved to PDF and is available in the repository for this assignment.

- AI tools were used in concept clarification and coding support; the report text was written without AI-generated prose.